

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Technical Presentation

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522



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COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 50

Code of Conduct

I A.LAKSHMI(192373066) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: LAKSHMI

Assessment Tasks

Technical Presentation Topics

Prepare a 6-8 minute presentation on any one of the following topics. Your slides must include definitions, architecture, industry relevance, and one practical example.

1. Cloud computing characteristics and benefits for startup ecosystems.
2. How virtualization enabled the growth of cloud service providers.
3. A comparative presentation on IaaS, PaaS, and SaaS for an educational institution.
4. Web services in cloud platforms: SOAP vs REST for enterprise integration.
5. Why ~~XaaS~~ is becoming a dominant consumption model in modern IT.

Minimum slide flow:

- Title and objective
- Core technical explanation
- Architecture or process diagram
- Real-world use case
- Conclusion and key takeaway



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Saveetha Institute of Medical And Technical Sciences
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CSA1522-Cloud Computing and Big Data Analytics

A Comparative Analysis of IaaS, PaaS, and SaaS for Educational Institutions

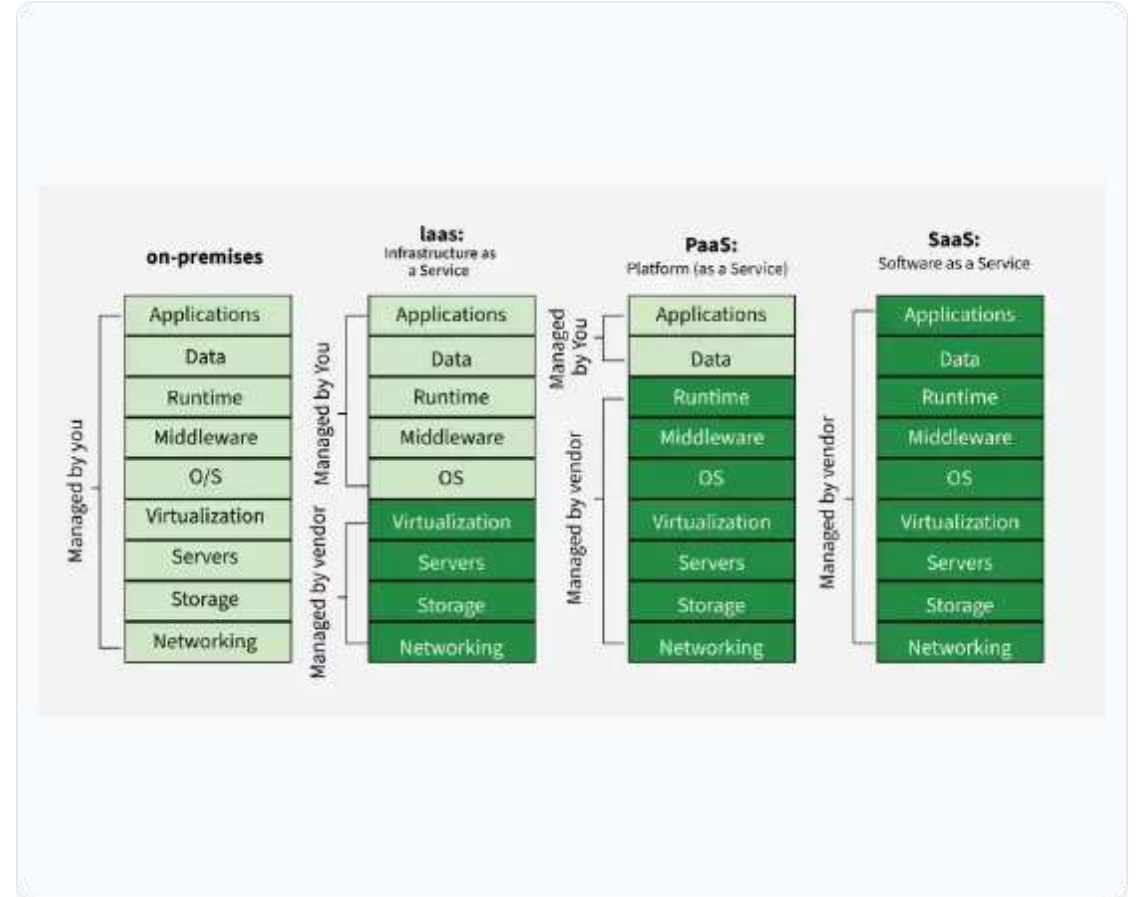
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The Cloud Control Spectrum

The primary difference between cloud service models lies in the division of responsibility between the institution and the cloud provider.

- **On-Premises:** The institution manages the entire stack, from physical hardware to applications. High control, but high capital and maintenance costs.
- **Infrastructure as a Service (IaaS):** The provider manages physical hardware and virtualization. You manage OS, data, and applications.
- **Platform as a Service (PaaS):** The provider handles infrastructure and OS/runtime environments. You focus on application code and data.
- **Software as a Service (SaaS):** The provider manages the entire stack. You simply consume the application as a user.



Defining the Core Cloud Models



IaaS

Infrastructure as a Service

Provides foundational computing resources (virtual machines, networking, storage) on demand over the internet.



PaaS

Platform as a Service

Delivers a managed framework for developers to build, test, and deploy custom applications without infrastructure headaches.



SaaS

Software as a Service

Delivers fully managed, ready-to-use software applications over the internet, typically accessed via a web browser.



IaaS in Action

Research Computing Environments

The Challenge: A university physics department needs massive computing power for a short-term data analysis project, but buying physical servers is cost-prohibitive.

The IaaS Solution: The IT team provisions high-performance Virtual Machines (VMs) and block storage on an IaaS platform (like AWS or Azure). The researchers install their specialized OS and custom analysis software.

The Benefit: Complete control over the specific environment required, rapid provisioning, and they only pay for the compute time used during the research phase.

PaaS in Action

Custom Student Portal Development

The Challenge: The university wants to build a unique, custom mobile application for students to track shuttle buses and check dining hall menus, but lacks a large IT operations team.

The PaaS Solution: Student developers and IT staff use a PaaS environment (like Heroku or Google App Engine). They write the code and connect to a managed database provided by the platform.

The Benefit: Faster time-to-market. The platform automatically handles security patching of the OS, load balancing during high traffic times (like lunch), and infrastructure management.





SaaS in Action

Learning Management Systems

The Challenge: Providing a robust, always-available platform for course materials, grading, and student communication across the entire university.

The SaaS Solution: The university subscribes to a cloud-hosted Learning Management System (LMS) like Canvas or Blackboard.

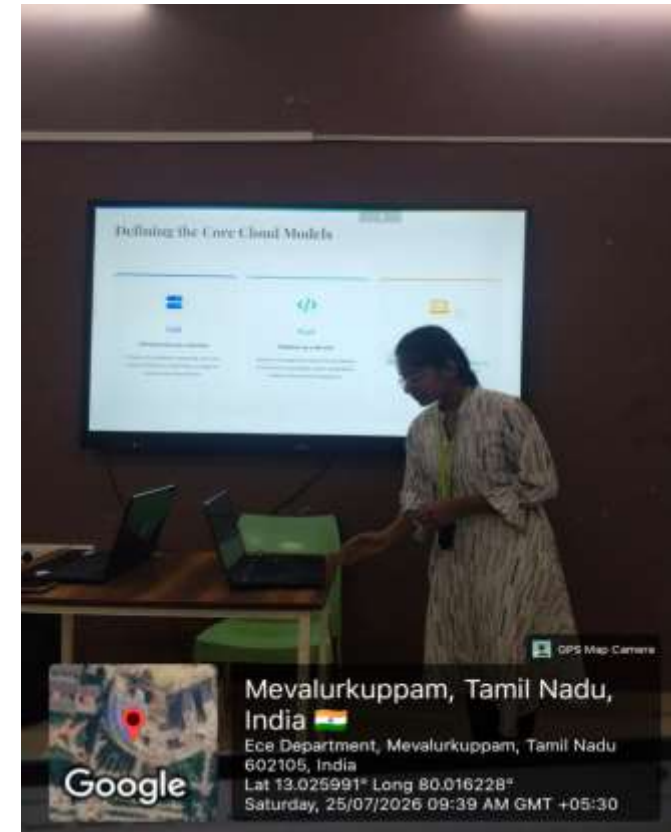
The Benefit: Zero infrastructure to maintain. The provider handles all software updates, security patches, backups, and scaling during peak exam periods. Users simply log in and use the tools.

Summary: Choosing the Right Model

Feature	IaaS (Infrastructure)	PaaS (Platform)	SaaS (Software)
Primary User	IT Administrators, Network Architects	Software Developers	End Users (Students, Faculty, Staff)
Level of Control	High (OS, Network, Data, Apps)	Medium (Data, Apps)	Low (User Access, App Configuration)
Management Burden	High (Requires specialized IT staff)	Medium (Focus on coding)	Low (Managed by vendor)
Time to Value	Days to Weeks (Configuration required)	Hours to Days (Focus on deployment)	Immediate (Ready to use)
Educational Examples	Hosting legacy student information databases	Developing a custom campus navigation app	Email (Google Workspace), LMS (Canvas)



CO1-AT3-PERSENTATION VIDOES



CO1-AT3-GPS PHOTO

Thank You